Title:ScriptBridge-Real-Time Offline Indic Script Transliteration Tool for Street Signboards

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github repository:
GitHub - NAKKA-POOJITHA/ScriptBridge: Real-Time Offline Indic Script Transliteration To...

deployed link: https://script-bridge-flame.vercel.app

problem statement: India is a linguistically diverse nation where a change in state borders often presents a completely new script. Travelers, pilgrims, and tourists frequently struggle to read critical geographic indicators like street signs and highway boards due to their inability to read the local script, creating severe navigation and accessibility barriers.

ABSTRACT

India is home to more than twenty officially recognized languages and over a dozen writing systems. While linguistic diversity enriches the country's cultural identity, it also creates practical barriers for interstate travelers, tourists, pilgrims, emergency responders, and migrant workers. Road signs, railway station boards, public safety instructions, and government notices are frequently displayed only in regional scripts, making them difficult to read for individuals unfamiliar with that script.

ScriptBridge is an offline mobile application designed to bridge this accessibility gap through real-time script transliteration. Instead of translating the meaning of the text, ScriptBridge converts the visual representation of one Indic script into another while preserving pronunciation. The application captures street signs through the smartphone camera, performs Optical Character Recognition (OCR), detects the source script, transliterates the extracted text into the user's preferred script, and overlays the converted text on the camera preview.

The system is built using Flutter for cross-platform mobile development, Google ML Kit Text Recognition for OCR, AI4Bharat Aksharantar/Indic Transliteration libraries for phonetic conversion, FastAPI for modular backend services during development, SQLite/Hive for local caching, and Flutter Text-to-Speech for audio accessibility. A major strength of ScriptBridge is its ability to execute core functionality entirely offline, enabling reliable operation in rural highways, remote villages, and low-connectivity environments.

The proposed solution demonstrates that lightweight edge-AI technologies can significantly improve navigation accessibility while maintaining privacy, low latency, and independence from cloud infrastructure.

Keywords: OCR, Transliteration, Indic Languages, Computer Vision, Offline AI, Mobile Computing, Accessibility, Flutter

1. INTRODUCTION

India is one of the most linguistically diverse countries in the world, with **22 officially recognized languages** written using multiple scripts such as Devanagari, Telugu, Tamil, Kannada, Malayalam, Bengali, Gujarati, Gurmukhi, and Odia. Road signs, railway station boards, public notices, hospital directions, and tourist information are generally displayed in the regional script of a particular state. While local residents can easily read these signboards, visitors from other states often face difficulties because they cannot interpret unfamiliar writing systems.

This problem is particularly significant for tourists, pilgrims, interstate travelers, migrant workers, and emergency responders, where the ability to quickly read destination names is essential for navigation and safety.

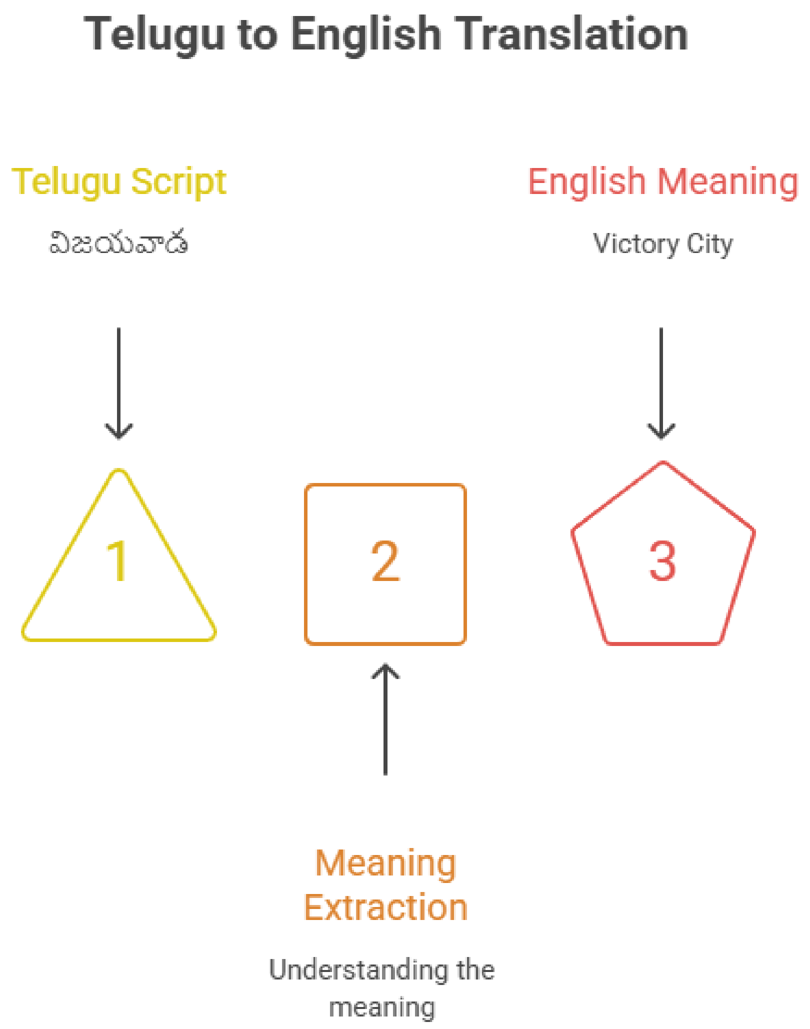
Most existing mobile applications focus on **language translation**, which converts the meaning of text from one language into another. However, for navigation purposes, understanding the meaning of a place name is usually unnecessary. Instead, users simply need the name written in a script they can pronounce.

Translation vs. Transliteration

Consider the Telugu street sign:

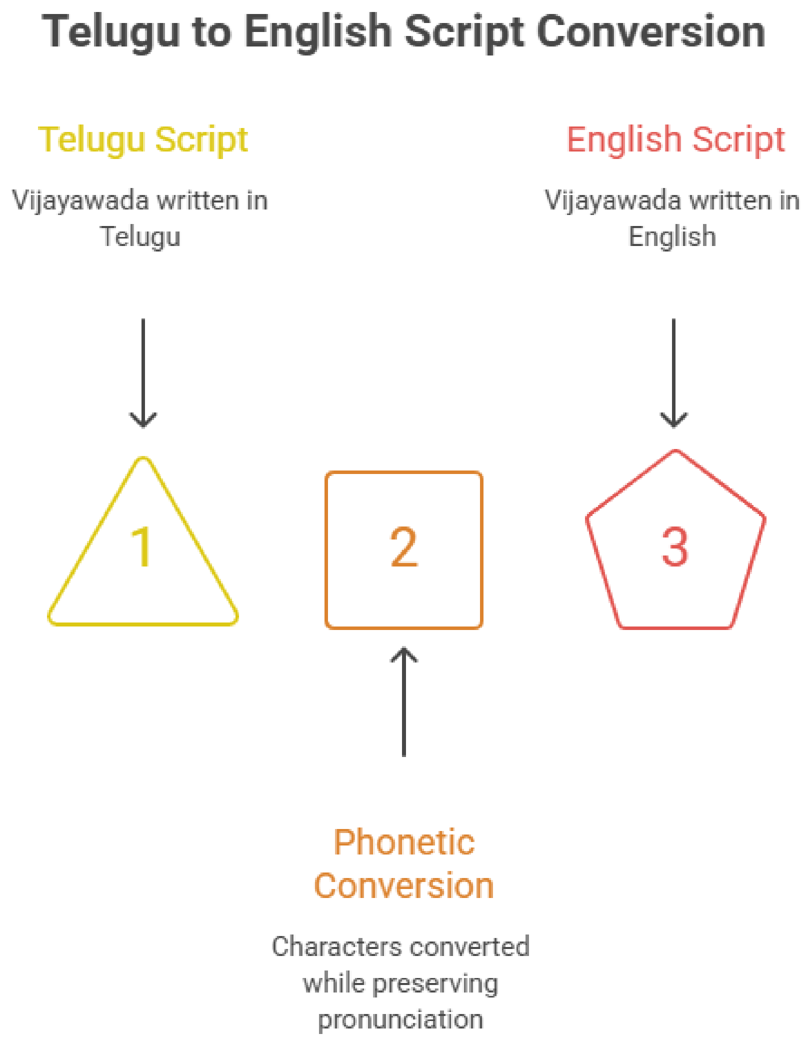
విజయవాడ

Translation (Meaning Conversion)



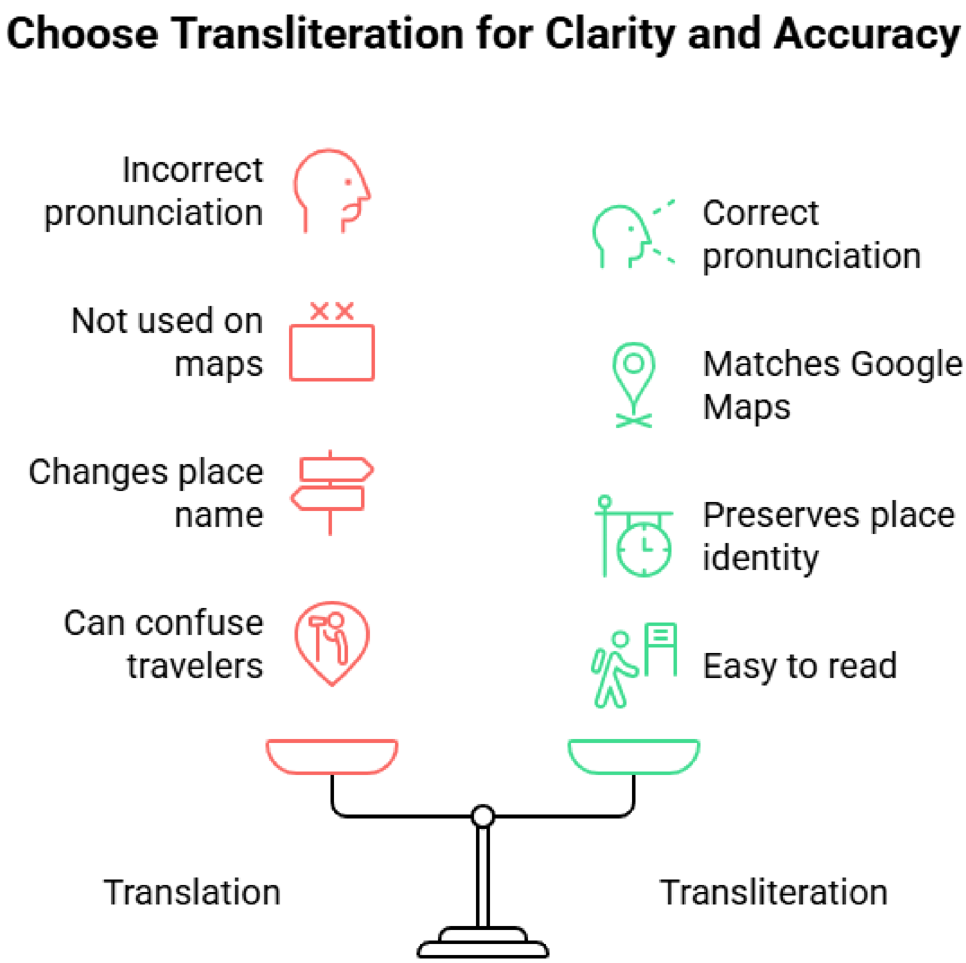
Problem:
The translated output changes the original place name and cannot be used directly for navigation, maps, or transportation systems.

Transliteration (Script Conversion)



Advantage:
The pronunciation remains exactly the same while only the writing system changes, enabling users to correctly read and pronounce the destination without altering its linguistic identity.

Why Transliteration is Better for Navigation



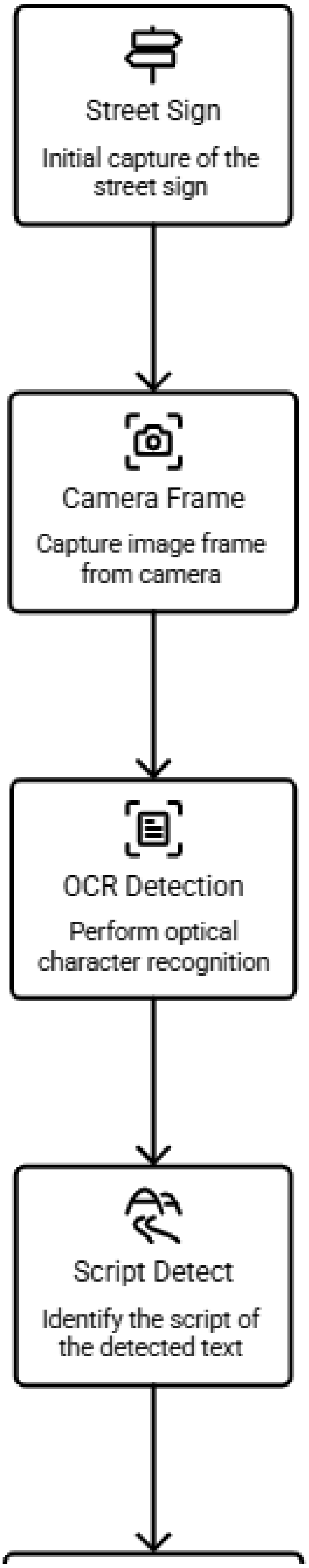
Proposed Solution: ScriptBridge

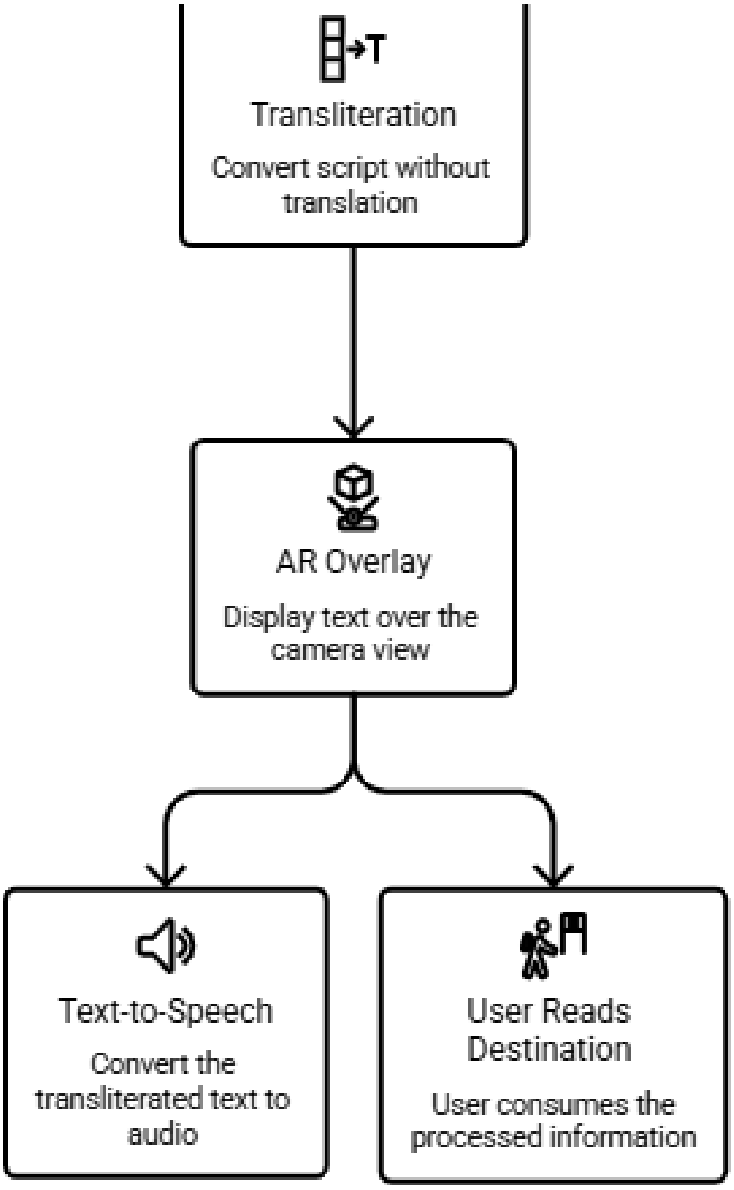
ScriptBridge addresses this challenge through **real-time offline script transliteration**. The application captures live images using the smartphone camera, extracts printed text using **Optical Character Recognition (OCR)**, detects the source script, performs phonetic transliteration into the user's preferred script, and overlays the converted text directly on the camera preview. The application also provides **Text-to-Speech (TTS)** functionality to read the transliterated text aloud, improving accessibility for visually impaired users and travelers.

Unlike cloud-based translation applications, ScriptBridge performs OCR, script detection, transliteration, and speech synthesis **entirely on the user's device**, ensuring low latency, enhanced privacy, and uninterrupted functionality even in remote areas without internet connectivity.

Overall Working of ScriptBridge

Street Sign AR Processing Workflow





2. REQUIREMENTS

2.1 Functional Requirements

The system shall:

- Capture live video from the mobile camera.
- Detect printed text in real time.
- Recognize multiple Indian scripts.
- Automatically identify source language/script.
- Transliterate extracted text into the selected script.
- Overlay transliterated text over the original signboard.
- Store scan history locally.
- Provide text-to-speech output.
- Operate completely offline.
- Allow users to select source and target scripts.

- Support both simulated demo mode and live camera mode.

2.2 Non-Functional Requirements

Performance

- OCR latency below 500 ms
- Smooth live camera preview
- Low CPU usage
- Fast transliteration

Reliability

- Works without internet
- Handles partial OCR failures
- Stable camera pipeline

Scalability

Future support for:

- 22+ Indian languages
- Handwritten OCR
- AR overlay enhancements

Security

- No cloud upload
- Local processing
- User privacy preserved

2.3 Hardware Requirements

- Android smartphone
- Camera
- 3 GB RAM minimum
- Quad-core processor
- Speaker

2.4 Software Requirements

Component	Technology
Mobile App	Flutter
OCR	Google ML Kit
Transliteration	AI4Bharat Aksharantar
Backend	FastAPI
Local Storage	SQLite/Hive
TTS	Flutter TTS
IDE	VS Code / Android Studio
Programming Language	Dart, Python

3. EXISTING SYSTEM

Several camera-based Optical Character Recognition (OCR) applications are available today, with **Google Lens** being one of the most widely used. Google Lens can recognize printed text from images, copy text, search for information, and translate text between languages. However, its primary objective is **language translation** rather than **script transliteration**.

For navigation applications, users generally do not require the meaning of a street name; instead, they need the original place name displayed in a script they can read. For example, a Telugu signboard reading **"విజయవాడ"** should ideally appear as **"Vijayawada"**, preserving its pronunciation. Traditional translation systems focus on semantic conversion and are therefore not optimized for this use case.

Furthermore, many existing solutions rely on cloud-based services for advanced translation, which can introduce latency in low-connectivity regions and may require users to upload image data to external servers. These systems also lack dedicated support for offline phonetic transliteration of Indic scripts and are not specifically designed for multilingual street-sign navigation.

Comparison of Existing System and Proposed System

Feature	Existing System (Google Lens)	Proposed System (ScriptBridge)
Primary Purpose	OCR and language translation	OCR with real-time script transliteration
Converts Meaning (Translation)	✓ Yes	✗ No
Converts Script (Transliteration)	Limited / Not dedicated	✓ Yes
Preserves Original Pronunciation	✗ Not guaranteed	✓ Yes
Designed for Street Sign Navigation	✗ No	✓ Yes
Real-Time Camera Processing	✓ Yes	✓ Yes
Offline OCR Support	Partial	✓ Yes
Offline Transliteration	✗ No	✓ Yes
Multi-Script Indic Support	Limited	✓ Optimized for Indian scripts
AR-Style Text Overlay	✗ No	✓ Yes
Text-to-Speech Accessibility	Limited	✓ Integrated
Scan History	✗ No	✓ Local Storage
Privacy	Some features may require cloud processing	Fully on-device processing

Limitations of Existing Systems

- Primarily designed for **language translation** rather than **script transliteration**.
- Do not preserve the original pronunciation of place names during navigation.
- Limited support for offline transliteration of Indian scripts.
- Some advanced features depend on internet connectivity or cloud services.
- No dedicated augmented reality overlay for multilingual street signboards.
- Limited accessibility features tailored for navigation, such as integrated pronunciation and offline speech assistance.

4. PROPOSED SYSTEM

ScriptBridge introduces a lightweight offline architecture for real-time script transliteration.

The workflow includes:

1. Camera captures street sign.
2. OCR extracts printed text.
3. Script detector identifies source writing system.
4. Transliteration engine converts script.
5. Overlay engine displays converted text.
6. Text-to-Speech reads pronunciation.
7. Local storage maintains scan history.

Unlike translation systems, ScriptBridge preserves pronunciation while changing only the writing system.

Example:

ಬೆಂಗಳೂರು

↓

Bengaluru

instead of

Bangalore

This distinction makes navigation more accurate.

Advantages

- Offline operation
- Fast response
- Low memory usage
- Privacy-preserving
- Cross-platform
- Supports multiple Indian scripts

- Accessible for visually impaired users

5. METHODOLOGY

The methodology consists of six major stages.

Stage 1: Image Acquisition

The application continuously receives frames from the mobile camera.

Captured frames are resized and optimized before OCR processing.

Stage 2: Optical Character Recognition

Google ML Kit Text Recognition extracts printed text.

Outputs include:

- Bounding boxes
- Confidence score
- Unicode text

Stage 3: Script Detection

Unicode ranges determine the detected writing system.

Example:

U+0C00–0C7F → Telugu

U+0C80–0CFF → Kannada

U+0900–097F → Hindi

U+0B80–0BFF → Tamil

Stage 4: Transliteration

The extracted text is passed into the transliteration engine.

Instead of translation,

తెలుగు

↓

Telugu

rather than

Language: Telugu

Pronunciation remains unchanged.

Stage 5: AR Overlay

The converted text is positioned using OCR bounding box coordinates.

The user sees

Original Sign

↓

Converted Script

in the camera preview.

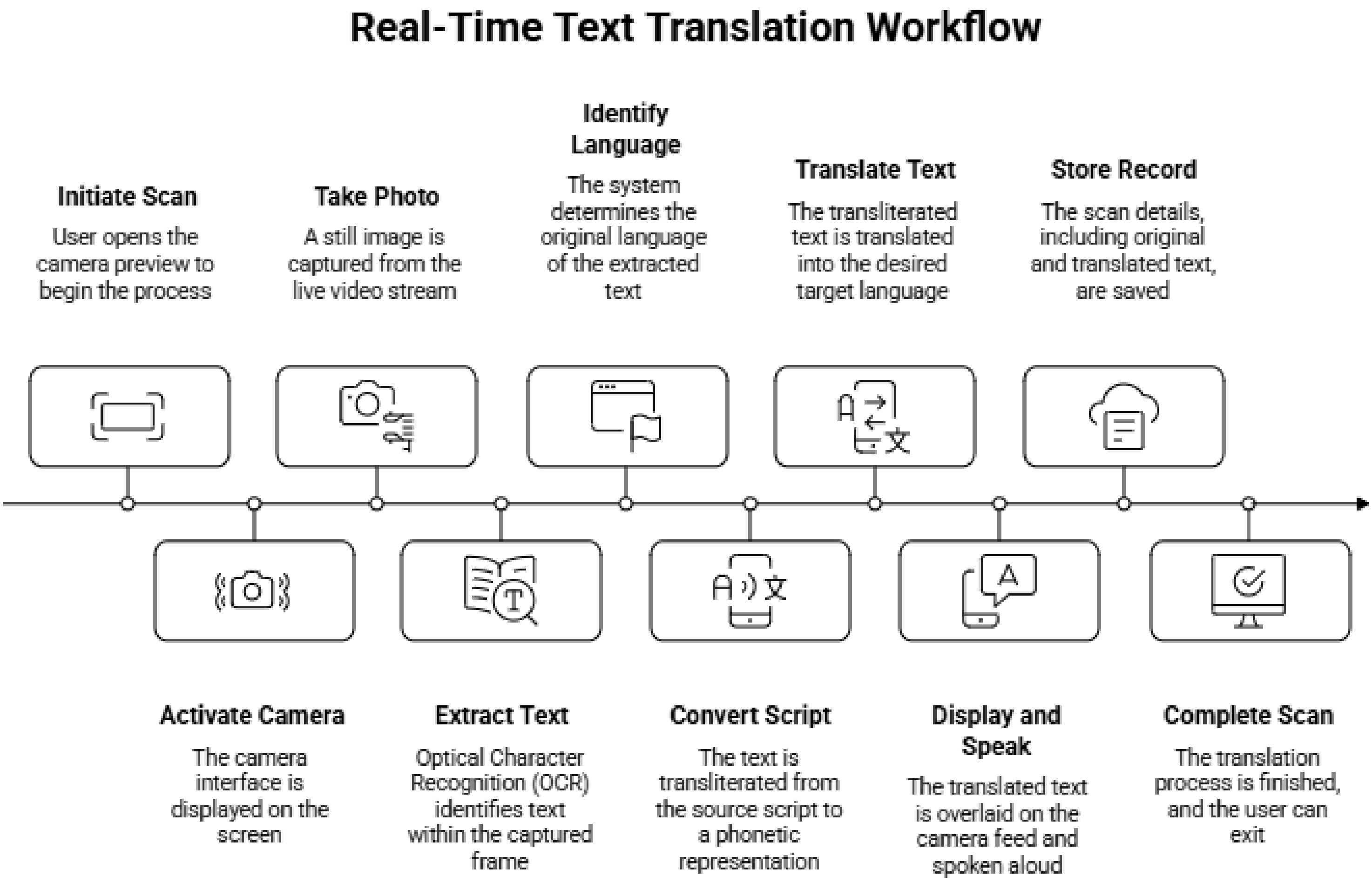
Stage 6: Audio Output

Flutter TTS reads the transliterated text aloud.

Speech parameters include:

- Speed
- Pitch
- Language selection

6. FLOW CHART OF METHODOLOGY



7. IMPLEMENTATION

Module 1

Camera Module

- Live preview
- Capture frames
- Frame throttling

Module 2

OCR Module

Google ML Kit detects

- words
- lines
- blocks

Confidence filtering improves recognition.

Module 3

Script Detection

Unicode inspection identifies:

- Telugu
- Kannada
- Tamil
- Hindi
- Malayalam
- Bengali
- Gujarati
- Punjabi

Module 4

Transliteration

Uses Indic transliteration mapping.

Supports

English ⇔ Telugu

English ⇔ Kannada

English ⇔ Tamil

English ⇔ Hindi

and other supported Indic scripts.

Module 5

Overlay Engine

Displays transliterated text using OCR coordinates.

Overlay updates continuously.

Module 6

Speech Module

Flutter TTS

Features:

- Auto speak
- Adjustable speed
- Adjustable pitch

Module 7

History

Stores

- source text
- transliteration
- timestamp

Offline database:

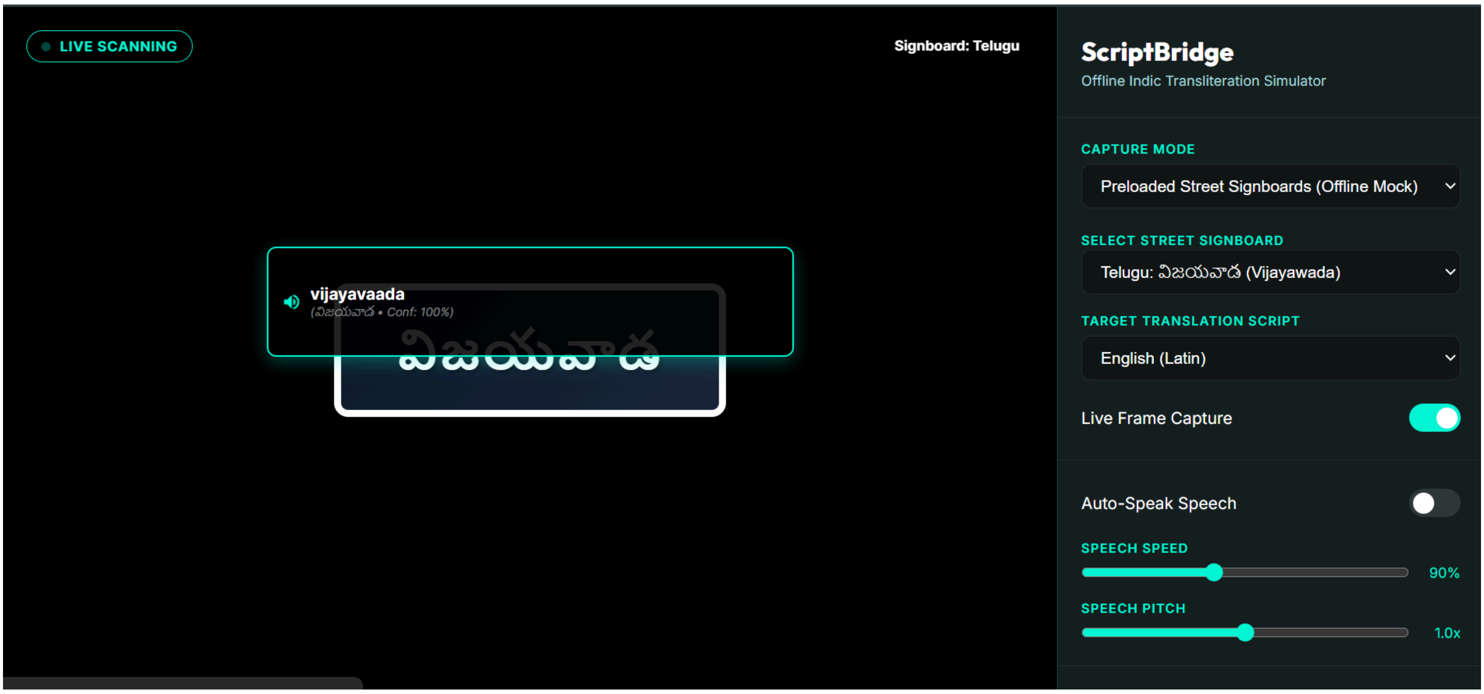
SQLite/Hive

8. RESULTS

The developed prototype successfully demonstrates real-time offline transliteration of multilingual Indian street signs.

Result 1

Live Camera Preview

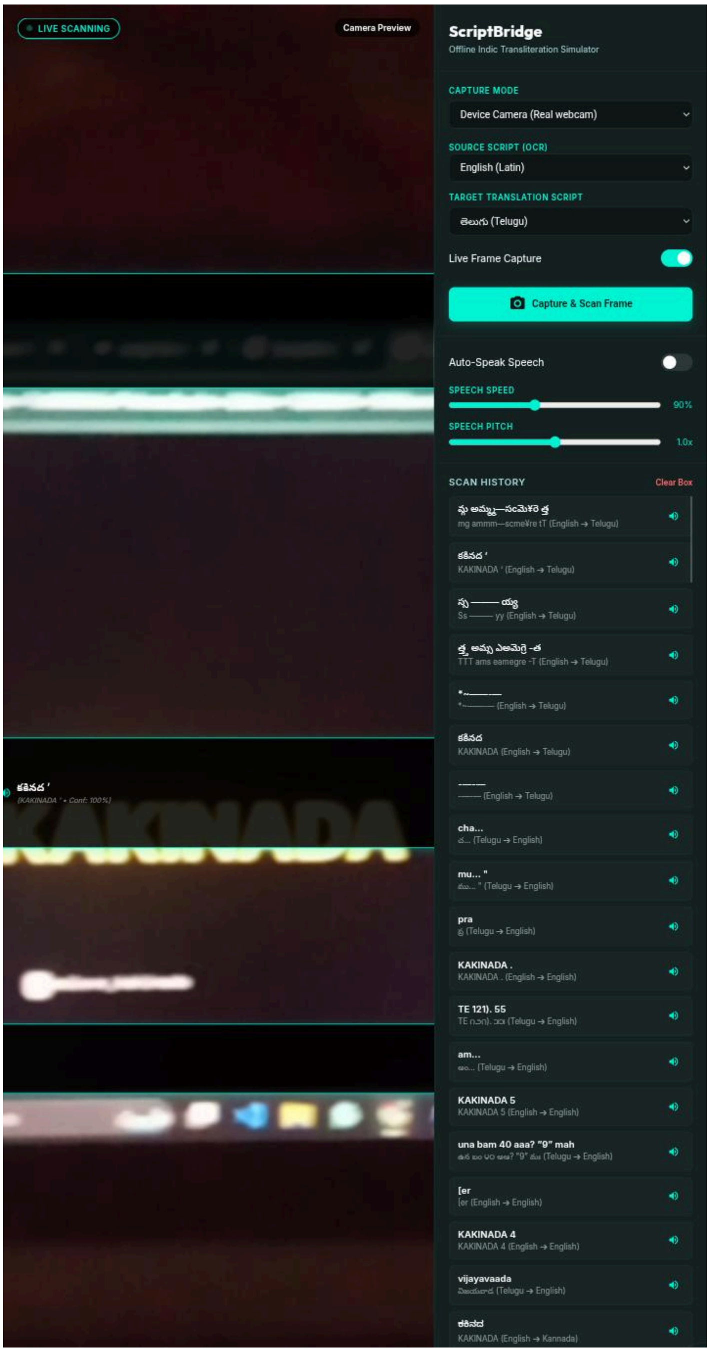


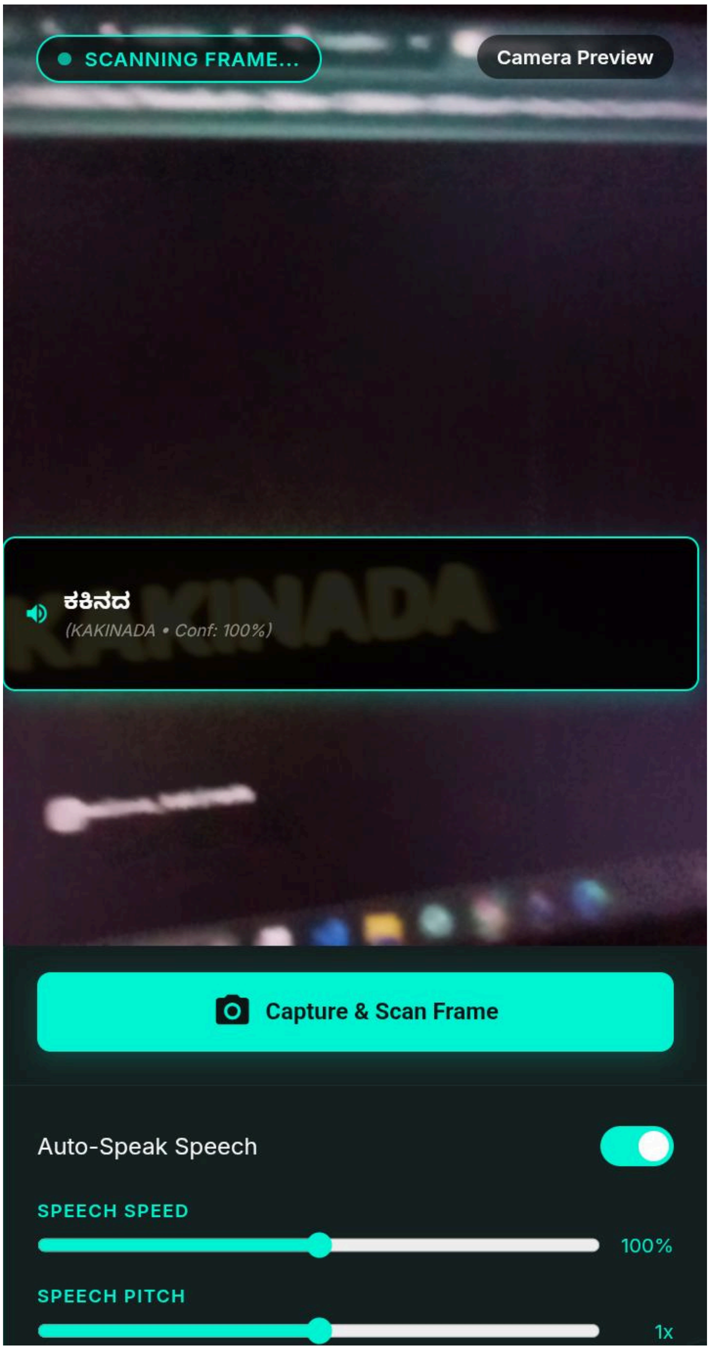
Features observed:

- Live scanning indicator
- Camera feed
- Real-time OCR

Result 2

Detected Text



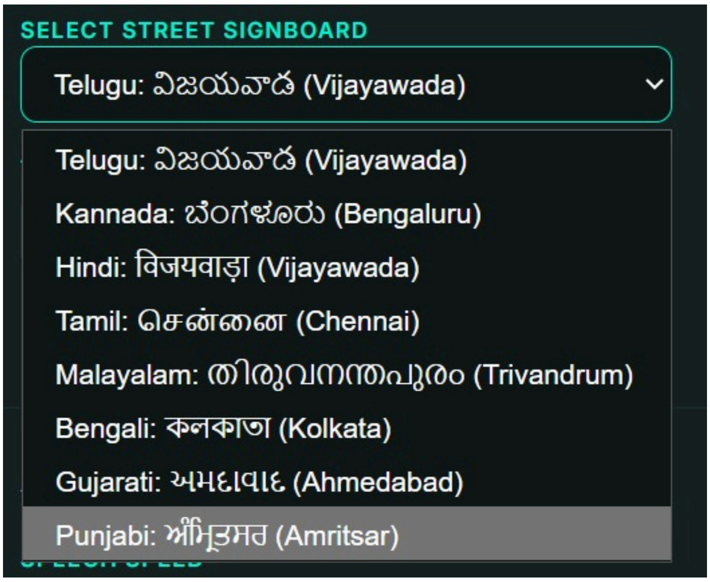


Displays

- confidence score
- recognized script
- transliterated output

Result 3

Language Selection

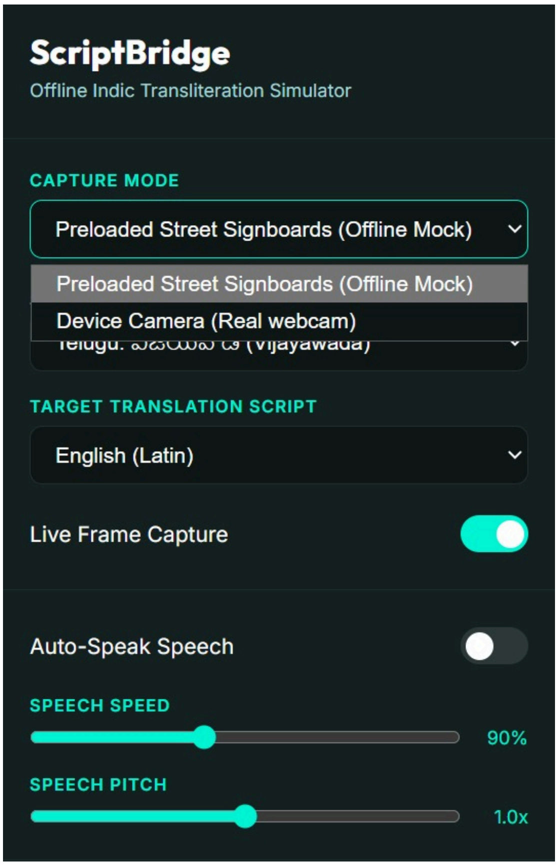


Supports

- English
- Hindi
- Telugu
- Tamil
- Kannada
- Malayalam
- Bengali
- Gujarati
- Punjabi

Result 4

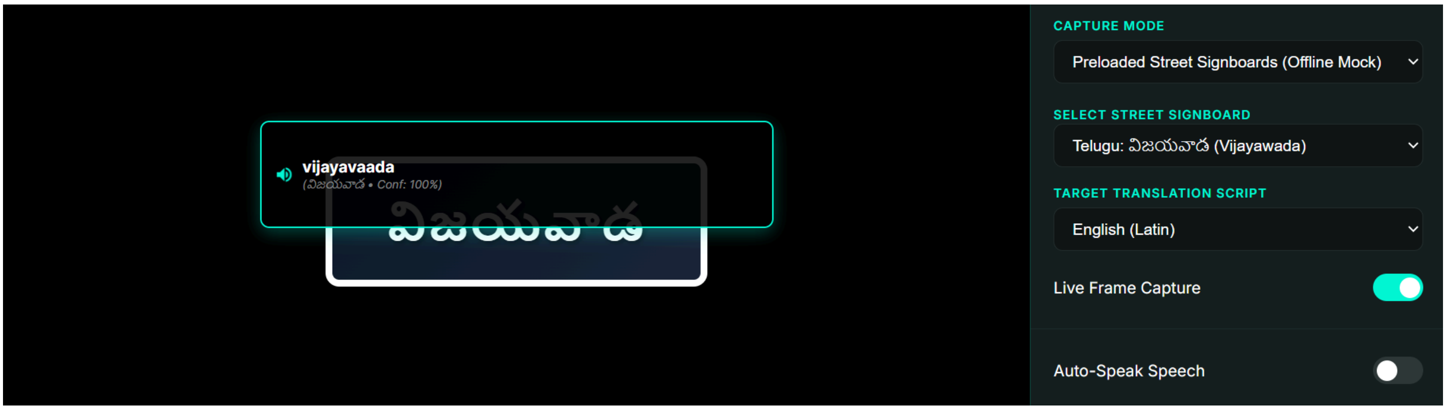
Offline Mode



Preloaded multilingual street signs.

Result 5

Speech Accessibility



Includes

- Auto Speak
- Speed adjustment
- Pitch adjustment

Observations

- OCR accuracy remains high for clear printed signs.
- Offline inference eliminates dependence on network connectivity.
- Latency is low enough for interactive use.
- AR-style overlay improves readability.
- Text-to-Speech enhances accessibility.

9. FUTURE WORK

Future enhancements include:

- Full ARCore-based spatial anchoring
- Handwritten text recognition
- Dynamic translation alongside transliteration
- GPS-assisted multilingual navigation
- Support for all 22 official Indian languages
- Integration with smart glasses and wearable devices
- Edge AI optimization using TensorFlow Lite
- Voice-controlled operation
- Automatic language detection using deep learning
- Cloud synchronization for scan history (optional)

10. CONCLUSION

ScriptBridge demonstrates an effective solution for overcoming script barriers in multilingual environments by combining OCR, script detection, transliteration, augmented reality overlays, and speech synthesis into a single offline mobile application. Unlike traditional translation systems, the application preserves pronunciation while converting text into a familiar writing system, enabling users to navigate confidently across linguistic regions without requiring internet access. Its privacy-preserving architecture, low latency, and accessibility-focused features make it especially suitable for travelers, pilgrims, emergency personnel, and residents in low-connectivity areas. With future enhancements such as expanded language support, advanced AR integration, and improved OCR capabilities, ScriptBridge has the potential to become a practical assistive technology for multilingual navigation across India.